



NEXT CHAPTER @ SMITH PUBLIC LIBRARY

Ratios Are Everywhere

Session 3 — Math Part B: Ratios & Percentage Applications

Monday Evenings • 5:30–7:30 PM • Library Conference Room

Tonight's Agenda

1

Quick recap of last week

10 min

2

Ratios in real life: concept

15 min

3

Ratio guided practice

25 min

4

Percentage applications: discounts, tips, tax

20 min

5

Two worked examples together

25 min

6

Mixed independent practice

20 min

7

Wrap-up & take-home preview

5 min

Quick Recap: Last Week

Order of operations:

Parentheses → Exponents → Multiply/Divide → Add/Subtract

Same number, three costumes:

$\frac{1}{2} = 0.5 = 50\%$ — fractions, decimals, and percentages are the same value in different forms.

Ratios Are Everywhere



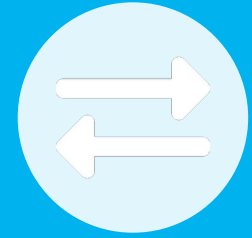
Recipes

“2 cups flour to 1 cup sugar” is a 2:1 ratio — double the recipe, keep the ratio.



Pay Rates

“\$18 for 2 hours” is a rate — the same idea as a ratio, applied to money and time.



Mixing & Scaling

Paint colors, cleaning solutions, and map scales all use ratios to stay proportional.

How to Solve a Ratio Problem

The Scale Factor Method

Step 1: Find what number the known side was multiplied by.

Step 2: Multiply the other side by that same number.

Example: A recipe uses 2 cups flour to 1 cup sugar. How much sugar for 6 cups flour?

2 was multiplied by 3 to get 6 → so multiply 1 by 3 too → 3 cups sugar.

Your Turn: Solve These Ratios

GUIDED PRACTICE

1

A recipe uses 3 cups flour to 2 cups sugar. How much sugar for 9 cups flour?

2

You earn \$45 for 3 hours of work. How much do you earn for 5 hours?

3

A map shows 1 inch = 10 miles. How many miles is 4.5 inches?

4

A paint mix is 2 parts blue to 5 parts white. How much white for 6 parts blue?

Let's Check Your Answers



1. 3:2 scaled by 3 $\rightarrow$ 9 cups flour needs 6 cups sugar



2. $\$45 \div 3 = \$15/\text{hour}$ $\rightarrow$ 5 hours = \$75



3. 1 in = 10 mi $\rightarrow 4.5 \times 10 = 45$ miles



4. 2:5 scaled by 3 $\rightarrow$ 6 parts blue needs 15 parts white

Percentages in Real Life

Discounts

“25% off” means you pay 75% of the original price.

Tips

A 20% tip on \$30 is $0.20 \times 30 = \$6$.

Sales Tax

8% tax on \$50 adds $0.08 \times 50 = \$4$, for a total of \$54.

Interest

Simple interest works the same way — a percentage of a starting amount.

Worked Example: Tip Calculation

Your restaurant bill is \$42. You want to leave an 18% tip. What's the total?

Step 1: Find 18% of \$42 $\rightarrow 0.18 \times 42 = \7.56

Step 2: Add the tip to the bill $\rightarrow \$42 + \$7.56 = \$49.56$

Total: \$49.56

Worked Example: Discount + Tax

A \$80 jacket is 30% off, then 8% sales tax is added. What's the final price?

Step 1: Find the discount $\rightarrow 0.30 \times 80 = \24

Step 2: Sale price $\rightarrow \$80 - \$24 = \$56$

Step 3: Find the tax $\rightarrow 0.08 \times 56 = \4.48

Step 4: Final total $\rightarrow \$56 + \$4.48 = \$60.48$

Final price: \$60.48

Independent Practice: Mixed Review

ON YOUR OWN

1

$$18 - 4 \times 3 = ?$$

2

Write $\frac{7}{10}$ as a percentage.

3

A recipe uses 4 cups water to 3 cups rice. How much water for 12 cups rice?

4

A \$25 shirt is 20% off. What's the sale price?

5

You tip 15% on a \$60 bill. How much is the tip?

Mixed Review Answers



1. $18 - 4 \times 3 = 18 - 12 = 6$



2. $7/10 = 0.7 \rightarrow 70\%$



3. 4:3 scaled by 4 $\rightarrow$ 12 cups rice needs 16 cups water



4. 20% of $\$25 = \$5 \rightarrow$ sale price $\$20$



5. 15% of $\$60 = \9 tip



Math Block: Complete.

See you next Monday — we build on this with algebra.

Before next week:

Try tonight's take-home practice. Want more? GED.com's free test preview tutorials (tools.ged.com) are always open, in English or Spanish.